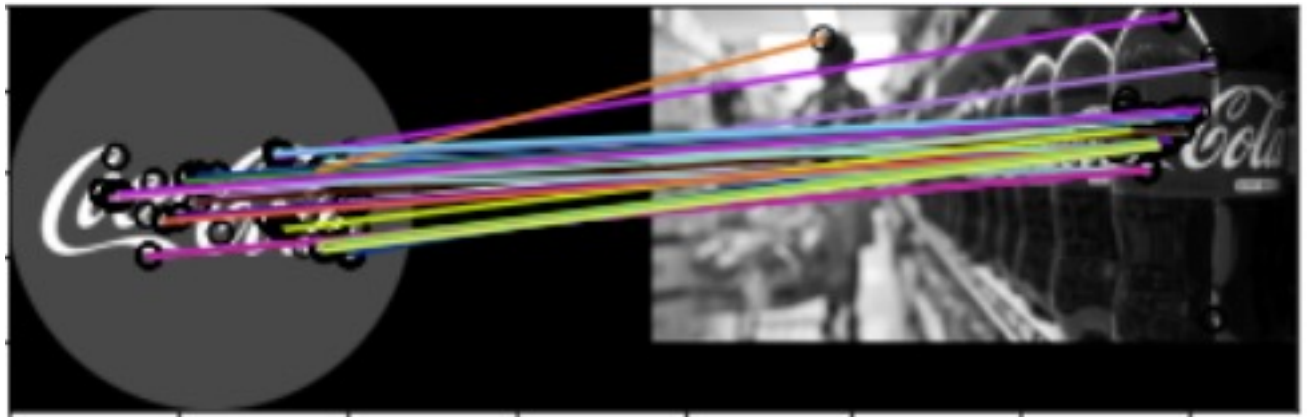
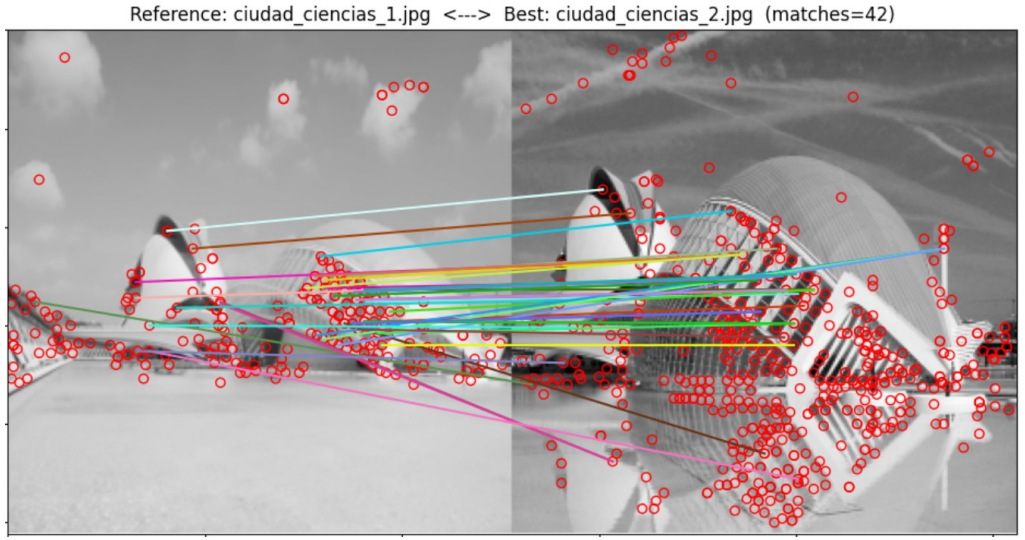
Instructions for Set 5

1. Corner detection (skimage.feature) and (fast, harris) modifiable threshold for peaks
2. Descriptor extractor (ORB, SIFT) and visualize keypoints
3. Matching with second image
4. Lookup closest image from folder



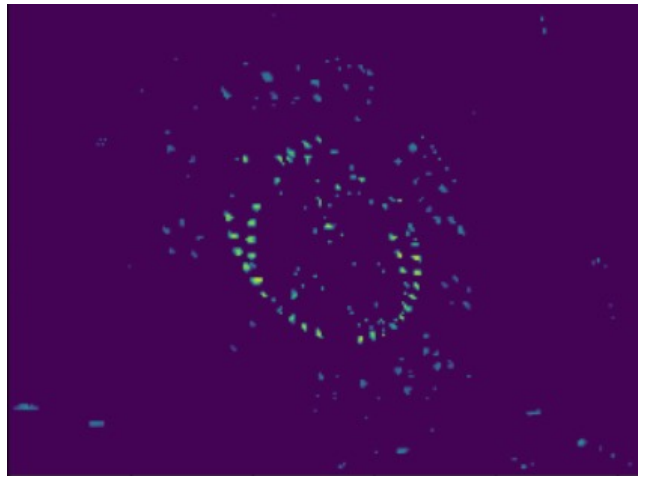




Set 5.1 Corner detection (skimage.feature) and (FAST, Harris) modifiable threshold for peaks

```
img = color.rgb2gray(img)

fast= feature.corner_fast(img)
plt.imshow(fast)
plt.show()
```



```
corner_points = feature.corner_peaks(fast)
plt.imshow(img, cmap=plt.cm.gray)
plt.scatter(corner_points[:, 1], corner_points[:, 0], facecolors='none', edgecolors='r')
plt.show()
```



```
harris_corners= feature.corner_harris(img)

corner_points = feature.corner_peaks(harris_corners)
```

Set 5.2 Descriptor and keypoints (ORB, SIFT)

```
descriptor_extractor = feature.ORB(n_keypoints=200)
```

```
descriptor_extractor.detect_and_extract(img1)
keypoints1 = descriptor_extractor.keypoints
descriptors1 = descriptor_extractor.descriptors
```

Must be configurable

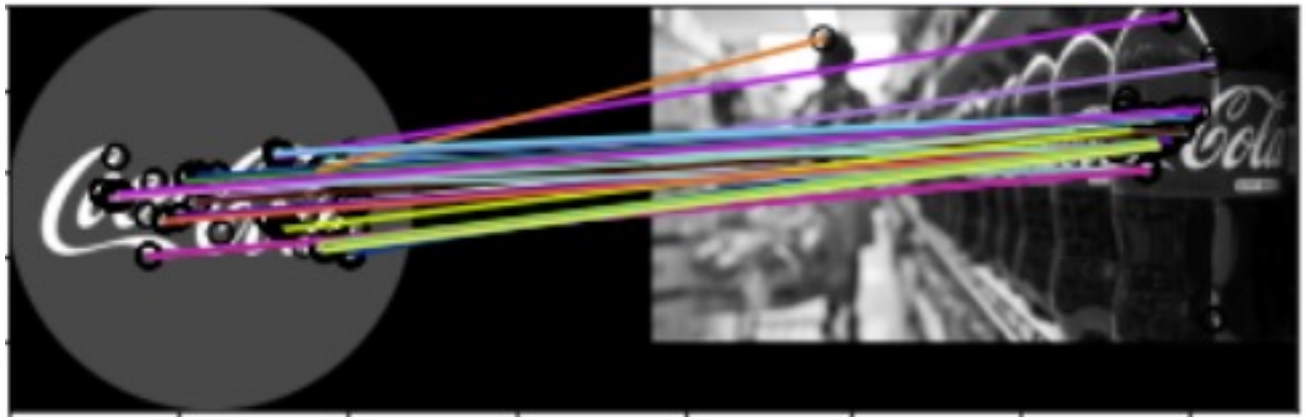
```
from skimage.feature import SIFT
```

```
sift_ref = SIFT(n_octaves=n_octaves, n_scales=n_scales, sigma_min=sigma)
sift_ref.detect_and_extract(ref_proc)
kp_ref = sift_ref.keypoints
desc_ref = sift_ref.descriptors
```



Set 5.3 Matchin with second image

```
matches = match_descriptors(desc1, desc2,
                            cross_check=cross_check,
                            max_ratio=max_ratio,
                            metric='hamming')
```



Set 5.4 Lookup closest image from a folder

The simplest algorithm at least (a better algorithm is welcome):
”return the image with maximum number of matches”

